Assignment 3

Due date: Monday, November 9, 2026, at 12:30 p.m. This assignment is worth 8% of the grade.

Choose Exercise 1 or Exercise 2. Indicate your choice in your report.

If used, clearly document in your report and code where you have made use of chatbots (like ChatGPT and so on...)

Exercise 1

Calibrate the local volatility model to the STOXX50E index using the Andreasen-Huge algorithm and the data provided. The resulting surface must contain implied vols for each expiration in the data, for strikes ranging from 40% to 200%. Take care to ensure that your implied vols match as closely as possible the market data in the spreadsheet; that is, your implied vols should interpolate the market data.

First derive the implied vols for the expirations in the spreadsheet, and then for the expirations $T = 1$ and $T = 1.5$. The market data (taken by the original Andreasen-Huge paper) refers to March 2010, when interest rates were very close to zero, and the dividend rate was also very low. Therefore, you can set $r = q = 0$.

Suppose that the stock follows a process

$$dS_t = S_t \sigma(S_t, t) dW_t \tag{1}$$

with

$$\sigma(S_t, t) = \frac{\tilde{\sigma}(S_t, t)}{S_t} \tag{2}$$

and $\tilde{\sigma}$ is a piecewise-constant function in S_t and t. Follow the steps in the paper and/or in the slides.

Notes: You can set the initial value of the index to $S_0 = 100$ (another initial value would however not affect results). Your report must include figures showing the volatility smiles (for each expiration) as a function of strike. It is not enough to just show the surface, you must also show cross-sections.

Include one figure showing the implied volatility smiles for the expirations in the data, and a separate figure showing the expirations $T = 1$ and $T = 1.5$.

Exercise 2

Calibrate the dynamic SABR model

$$dF_t = \sigma_t F_t^{\beta} dW_t^F \quad (3)$$
$$d\sigma_t = \alpha_t \sigma_t dW_t^{\sigma} \quad (4)$$
$$dW_t^F dW_t^{\sigma} = \rho_t dt \quad (5)$$

with

$$\rho(t) = \rho_0 e^{-at} \quad (6)$$
$$\alpha(t) = \alpha_0 e^{-bt} \quad (7)$$

where $a > 0,\ b > 0$ are constants. See the slides for details as well as the provided paper: "Static and dynamic SABR stochastic volatility models" (arXiv:2407.20713).

Use the following table of prices for STOXX50 options (expressed in terms of implied volatility):

K (% of price)	σ_{mkt} (%) $T = 0.25$	σ_{mkt} (%) $T = 0.5$	σ_{mkt} (%) $T = 1$	σ_{mkt} (%) $T = 2$
0.88	32.21	32.07	30.11	28.16
1	29.79	29.63	28.22	26.54
1.12	27.52	27.42	26.34	24.95

After calibrating your model, you can use the model to compute option prices and thus implied vols for arbitrary strikes and expirations. Use this to make some nice figures!

In particular, your report must include at least one figure showing the cross-sectional implied vol smiles as a function of strike, for each expiration in the table. Ensure that you indicate the actual data on this figure, so that you show your fitting performance visually. It is not enough to just show a surface, you must also show cross-sections.